



IEEE TECHSANGAM 2025



- **Problem Statement Title- Addressing Delayed and Inefficient Fire Response with an Autonomous IoT-Based Long-Distance Fire Extinguishment System**
- **Theme- Smart City**
- **Domain- Internet of Things**
- **PS Category- Hardware and Software**
- **Team Name (As per Registration)- FireLinX**

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Problem Statement

The Critical Gap: Beyond Sprinklers & Hoses INDIA'S FIRE CRISIS IN NUMBERS:

OVERVIEW OF THE PROBLEM:

- 18,000+ deaths annually from fire incidents.
- 40% urban fires erupt in densely packed slums with no safety infrastructure.
- 24,000 forest fires rage yearly, threatening ecosystems.
- 20,000+ industrial incidents (2020) due to safety gaps.

WHY MINUTES MATTER !!

- Traditional firefighting faces deadly delays: Dispatch → Travel → Setup = Lost time.
- Sprinklers fail. Hoses can't reach. Hydrants are inaccessible.
- High-rises, slums, forests, factories = logistical nightmares.

The Unforgiving Minutes: When Fire Outpaces Hope !!

Urban slums lack infrastructure;
Factories face system failures;
Forests suffer detection delays.
High-rises struggle with access.

Existing solutions fail when sprinklers overwhelm or responders arrive too late, turning containable fires into disasters.

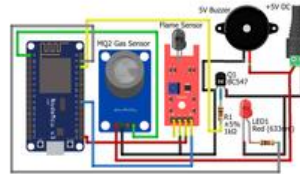
Critical Pain Points !!

1. Detection-to-action lag: Alerts trigger reactions, not solutions.
2. Static infrastructure: Fixed systems can't adapt to changing fire spread shifts.
3. Fragile coordination: Siloed sensors & suppressors delay unified action.

Why This Matters ?

Current systems remain reactive, not proactive—prioritizing alerts over action and stability over adaptability.

RESPONSE: FRAGMENTED & DELAYED ACTION



IoT sensors (e.g., Cambridge SmartFire)
Detect fires but lack automated suppression, relying on human intervention.



Predictive tools (e.g., NFPA frameworks)
Analyze risk but do not trigger real-time countermeasures, losing critical containment time.

READINESS: SYSTEMIC LIMITATIONS IN CRISIS MANAGEMENT

Manual Systems (e.g., Fire-trucks 🚒)

face issue like:

- 🚧 Urban gridlock; 🏠 Narrow slum lanes;
- 🚫 Remote access issue Causing delay in response by 15–45 mins (as per NFPA 2023 data).



Satellite-based detection ✨ (e.g., ESA's REMSAT II)

Suffers 15–30 min orbital latency 📶
Too slow for fast-spreading fires.



REACH: INADEQUATE COVERAGE & MOBILITY



Static installations (sprinklers, water cannons)
15–30 min orbital latency 📶
Too slow for fast-spreading fires.

Robotic systems (e.g., IIT Madras' indoor bots)

Lack terrain adaptability, struggling with rubble, slopes & dense smoke.



RESILIENCE: VULNERABILITY UNDER STRESS



Power/water-dependent systems (fire-hydrants, etc.)

Fail during outages or pipe-damage.

Human-dependent protocols (DRDO's surveillance robots)

Stress skews decisions, thus escalating risk.



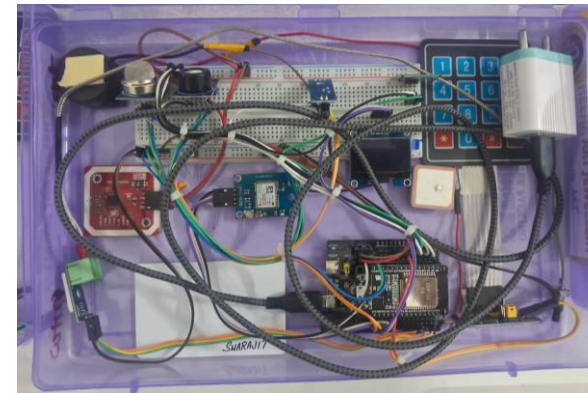
Our device FireLinX, an automated system, bridges the gap :

1. IoT-driven instant detection; 2. Long-range precision targeting; 3. Adaptive retardant extinguishment; 4. Web Remote- Control.



Intelligent Detection & Targeting:

- Real-time Multi-Sensors (Gas, Flame, Temp) + GPS Localization.
- Automated 3-Axis nozzle aims using calculated trajectory (Haversine/Bearing).

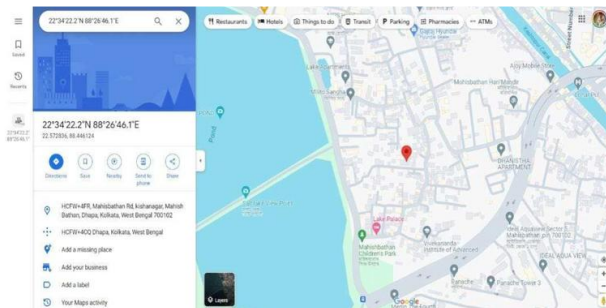


Remote Oversight & Control:

- Web Dashboard (Node.js/React) provides live data, map view.
- Dah also has Analytics & Secure manual override capabilities.

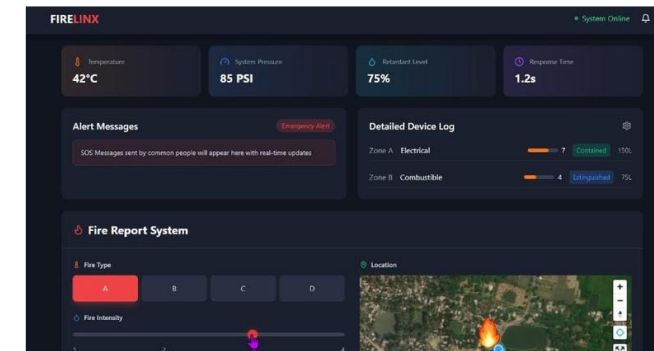
Resilient Communication:

- Tri-Channel system: RFID-Secured IoT Pads, Web/MQTT, SMS,)
- Ensures 24/7 operation even if 1 network fails.



Adaptive Suppression:

- Automatically selects correct retardant (Water/Foam/CO2) based on fire type.
- Dynamically adjusts pump speed (PWM) for optimal flow.



Technical Approach

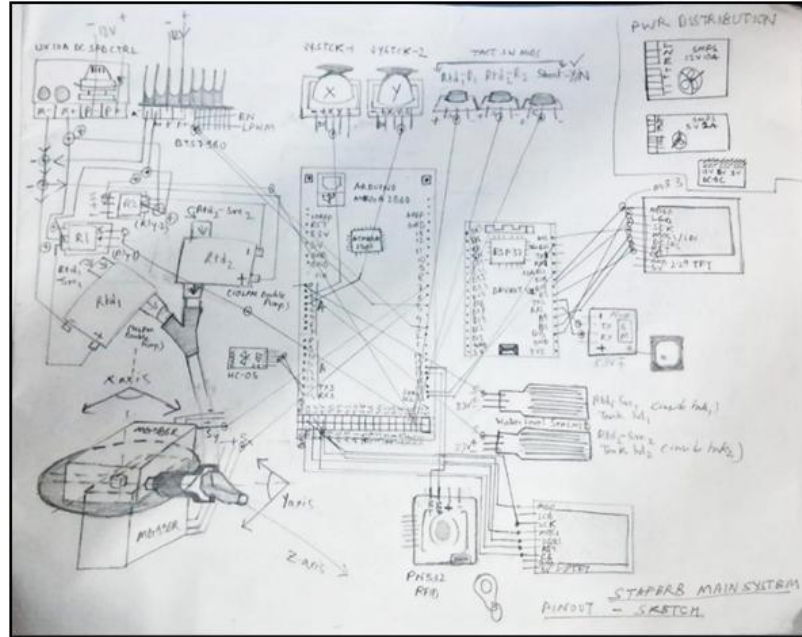
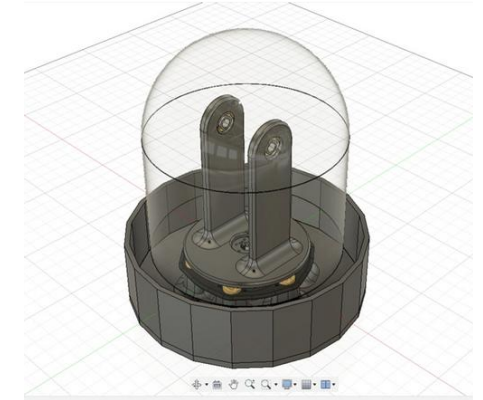


Fig. 1. Main Station Sketch

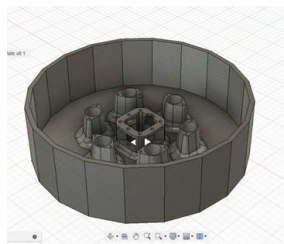
STAFERB Main Station Architecture (Refer to Fig.1 Schematic)

Core Components:

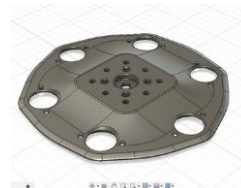
- **Processing Hub:**
 1. Arduino Mega (data handling) + ESP32 (IoT comms).
 2. GSM module for SMS alerts.
- **Suppression Unit:**
 1. 3-axis servo nozzle (X/Y/Z targeting).
 2. PWM-regulated pump (150–255 pressure range).
- **Power System:**
 1. 240V AC to 12V 15A DC SMPS.
 2. Weatherproof, corrosion-resistant chassis.
- **Suppression Logic:**
 1. **Retardant Selection:** Water (Class A) / Foam (Class B) / CO₂ (Class C).
 2. **Precision Targeting:** Haversine formula (distance) + Bearing formula (nozzle angle).
 3. **Dynamic Control:** Real-time sensor feedback adjusts flow rate.
- **Manual Operation:**
 1. **RFID-Auth Access:** Secure operator override.
 2. **Joystick Control:** Manual nozzle positioning.
 3. **Emergency Protocols:** Fail-safe activation.



Full prototype CAD. Dome will be opaque. Kept transparent for demonstration.



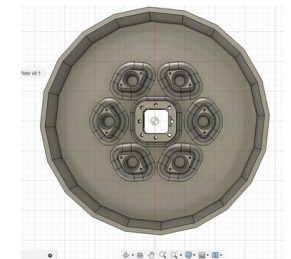
This is the base of the fire extinguisher unit. It will hold the stepper responsible for moving the gimble in x axis. It will also hold ball castor bearing so that the upper plate moves freely



This is the upper plate which will secure the ball castor bearings and base together. It will remain stationary with the base



This is the rotating part of the design which will hold the nozzle and tilt it in y axis. the bottom part is designed to move freely on the ball castor bearings.



Top view of the base

Note : The website images are for illustration purposes. These are designs of Canva. No actual website is being built yet.

Technical Approach

Detailed Proposal & Solution Approach: IoT-enabled Subsystem & Web-Dashboard



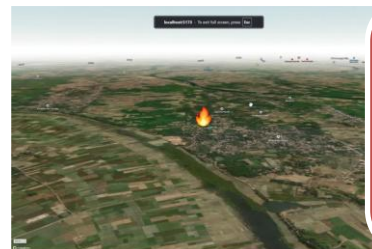
ESP32 Subsystem Manual Alerts Sent Parsing ESP32 Subsystem Sensor Circuitry (Gas, Flame, High-Temperature)



Fire-Type & Fire-Intensity Display Master Esp-32node ALERT reception Arduino MegaAlert Parsing



Landing Page
This section presents the project's core concept and key features, with the 'Get Started' button directing users to the Fire Report page.



Enhanced Map Feature
The enhanced mapping interface provides real-time geolocation tracking of fire incidents, with visual indicators displaying intensity levels and fire type classification.

ESP32-Based IoT Fire Detection & Alert Network

1. Automated Fire Detection (Ref: Fig.2 Sensor Node Schematic)

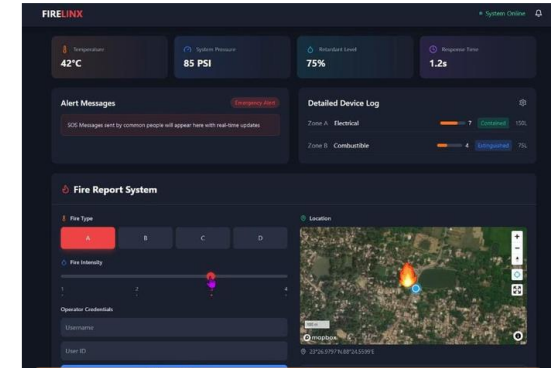
- Sensor Array:**
 - MQ-2/MQ-135 gas sensors for flammable vapor detection.
 - Flame sensors for heat anomaly analysis.
 - NEO-6M GPS for real-time incident coordinates.
- Environmental Monitoring:**
 - Continuous air quality checks.
 - Thermal imaging via infrared sensors.

2. Manual Fire Reporting

- RFID-Auth Protocol:**
 - Verified personnel access via EM-18 RFID readers.
 - OLED keypad for fire classification (Type A/B/C/D, Intensity 1–4).
- Alert Transmission:**
 - Structured SMS alerts: LOC:<coordinates> INT:<1-4> TYPE:<A/B/C/D>.
 - IoT-MQTT integration for real-time dashboard updates.

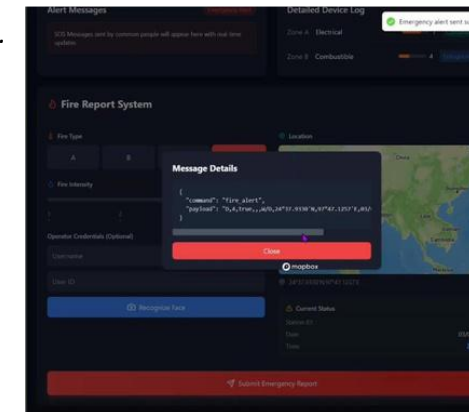
3. Key Features

- Redundant Communication:** GSM + IoT-MQTT ensures uptime.
- Low Latency:** Alerts processed in <5 seconds
- Compliance:** Aligns with NFPA 72 fire alarm standards.



Fire Report

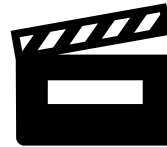
Users and authorized personnel can submit critical fire incident details including fire type, intensity level, name, and ID through the platform, which securely transmits this data to the online MQTT broker for real-time monitoring and response.



WEB to Master ESP
This module transmits fire report data to the master ESP32 device through an online MQTT server for centralized monitoring.

Tools and devices used on development

- Arduino Mega (Processing Hub)
- ESP32 (IoT Comms & Subsystem)
- GSM module (for SMS alerts)
- 3-axis servo nozzle (X/Y/Z targeting)
- PWM-regulated pump (12V DC Pump implied)
- SMPS (240V AC to 12V/15A)
- MQ-2/MQ-135 gas sensors
- Flame sensors
- NEO-6M GPS module
- OLED keypad/display
- EM-18 RFID readers
- Joystick (for Manual Control)
- CAD Software (Implied - for 3D models)
- 3D Printer (Implied - for custom parts like base, upper plate, rotating part)



Technologies involved/used

IoT (Internet of Things) Communication Protocols:

- Web-Dash & Station Integration: MQTT Protocol
- Redundant Communication channels for Subsystem: ESPNOW, UDP
- SMS Communication (Structured Alerts)

Embedded Systems Programming (Arduino/ESP32):

- Real-time Sensor Data Processing
- PWM, Servo Motor Control (for nozzle)
- Haversine Formula, Bearing Formula (for distance & angle calculation)
- RFID Authentication

Hardware Designing: 3D Modeling & CAD

Web Development:

1. **Frontend:** React, Tailwind CSS
2. **Backend:** NodeJS, ExpressJS
3. **Database:** MongoDB, Firebase
4. **AI/ML:** TensorFlow, PyTorch
5. **Geolocation & Mapping (for Dashboard)**

[Github link](#)

[Pre-build demo video](#)

[Demo video link for all communication channels of FireLinX](#)

Feasibility Analysis

FIRELINX (STAFERB) is a technically sound and innovative project designed to respond quickly to fire emergencies using IoT, GPS, and real-time automation. Its feasibility is backed by:

- 1. Proven Technology:** Utilizes widely used modules (Arduino, ESP32, sensors, GPS).
- 2. Multi-channel Communication:** Operates through GSM, IoT, and manual input—ensuring availability even during outages.
- 3. Practical Use Cases:** Easily adaptable to urban, industrial, slum, and forest environments.
- 4. Low Cost:** Components are inexpensive and locally available, ensuring economic scalability.
- 5. Modular & Scalable:** Can be deployed as standalone units or networked clusters.

Potential Challenges and Risks

- 1. Environmental Conditions**
Dust, heat, rain, or extreme weather can affect sensors or hardware.
- 2. Power Dependency**
Continuous operation needs a reliable battery backup or solar power, especially in remote areas.
- 3. Network Failures**
GSM or Wi-Fi signals may be weak or unavailable in forest or rural zones.
- 4. Hardware Wear & Tear**
Moving parts like servo motors and pumps may fail with prolonged use.
- 5. False Positives/Negatives**
Sensors might misinterpret signals, triggering unnecessary actions or failing to detect real threats.
- 6. Security Risks**
Unauthorized system access if RFID/IoT is compromised.

Strategies for Overcoming Challenges

- 1. Rugged Enclosure Design**
Use waterproof, heat-resistant, and dustproof casings to protect hardware.
- 2. Battery & Solar Backup**
Add solar charging units and larger battery packs to ensure uptime in off-grid areas.
- 3. Multi-Network Failover**
Include both GSM and LoRaWAN/Wi-Fi for redundancy in communication.
- 3. Scheduled Maintenance Protocols**
Implement predictive maintenance alerts for pumps, motors, and sensors.
- 4. AI-Powered Filtering**
Use machine learning models to distinguish real fires from noise (false positives).
- 5. Encrypted Access & Logs**
Encrypt RFID verification and use logs to track system access and misuse attempts.

Impact and Benefits



Urban & Residential:
Incl. Dense Slums &
Planned Colonies.



Industrial & Commercial:
Factories, Warehouses,
Chemical/Oil/Gas.



Remote & Environmental:
Forest Fire Management.



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Forest Fire Management.



Remote & Environmental:
Forest Fire
Management.

Official Infrastructure & Defense:
Power Plants, Data Centers,
Depots, Military Bases,
Borders.



Research and References

- NASA's FireSat Project: A satellite-based system designed to detect wildfires globally in real-time, aiding firefighters in quick and effective fire suppression efforts.
- University of Cambridge's "SmartFire" initiative: By integrating IoT sensors and data analytics to monitor and predict forest fires in real-time, it enhances earlywarning systems and enabling more efficient firefighting strategies.
- IIT Madras' Fire Extinguishing Robot: Autonomous robotic system for navigating indoor environments and extinguishing fires.
- ESA REMSAT II Project: Utilizes satellite data to monitor wildfires globally, providing critical information for early detection.
- Carnegie Mellon University's Multi-Robot Team: Develops autonomous robots for indoor fire exploration and suppression using ML.